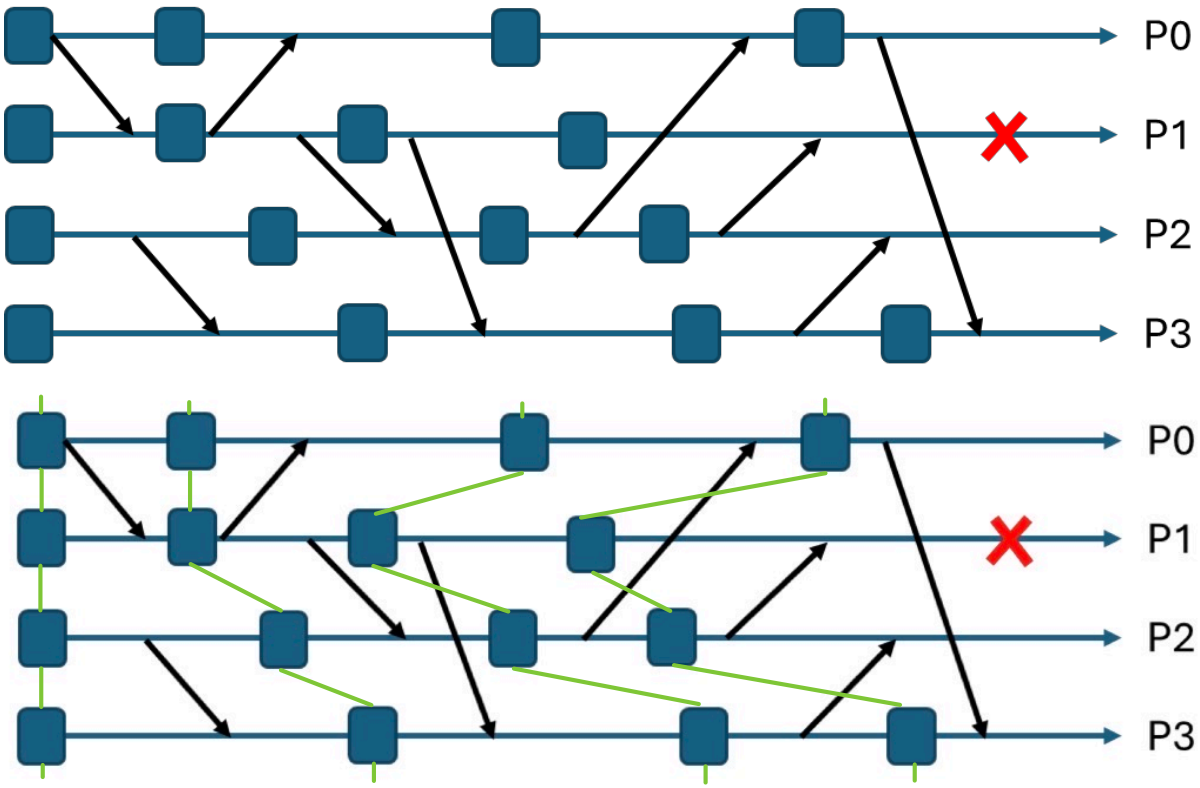
Exercise: Checkpointing
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The figure below shows the timeline of message-passing system which is distributed into 4 processes (P0-P3). The arrows show message passing from one to the another process. The **timestamped states** on each process are placed with the rectangle blocks. In this figure, the **process P1** once has faced to the **error** state at time **X (red).** In order to restore to the error-free state, draw the **consistent cuts** and **strongly consistent cuts** for all process to generate the recovery line.



Consistent cut : each green line drawn across check point block (P_0 to P_3) represent consistent cut and strong consistent cut because no message transmission arrows cross over these line in the way that create orphan message

Recovery line : The 4th green line serves as the recovery line